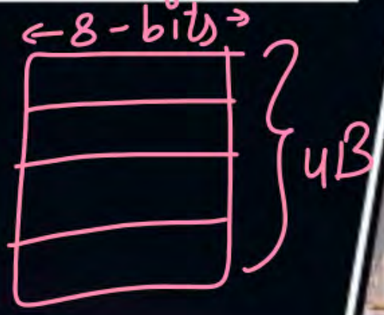
CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

IO Organization



Lecture No.- 3



By- Vishvadeep Gothi sir

Recap of Previous Lecture



Topic Interrupt Mode

Topic DMA

Topics to be Covered

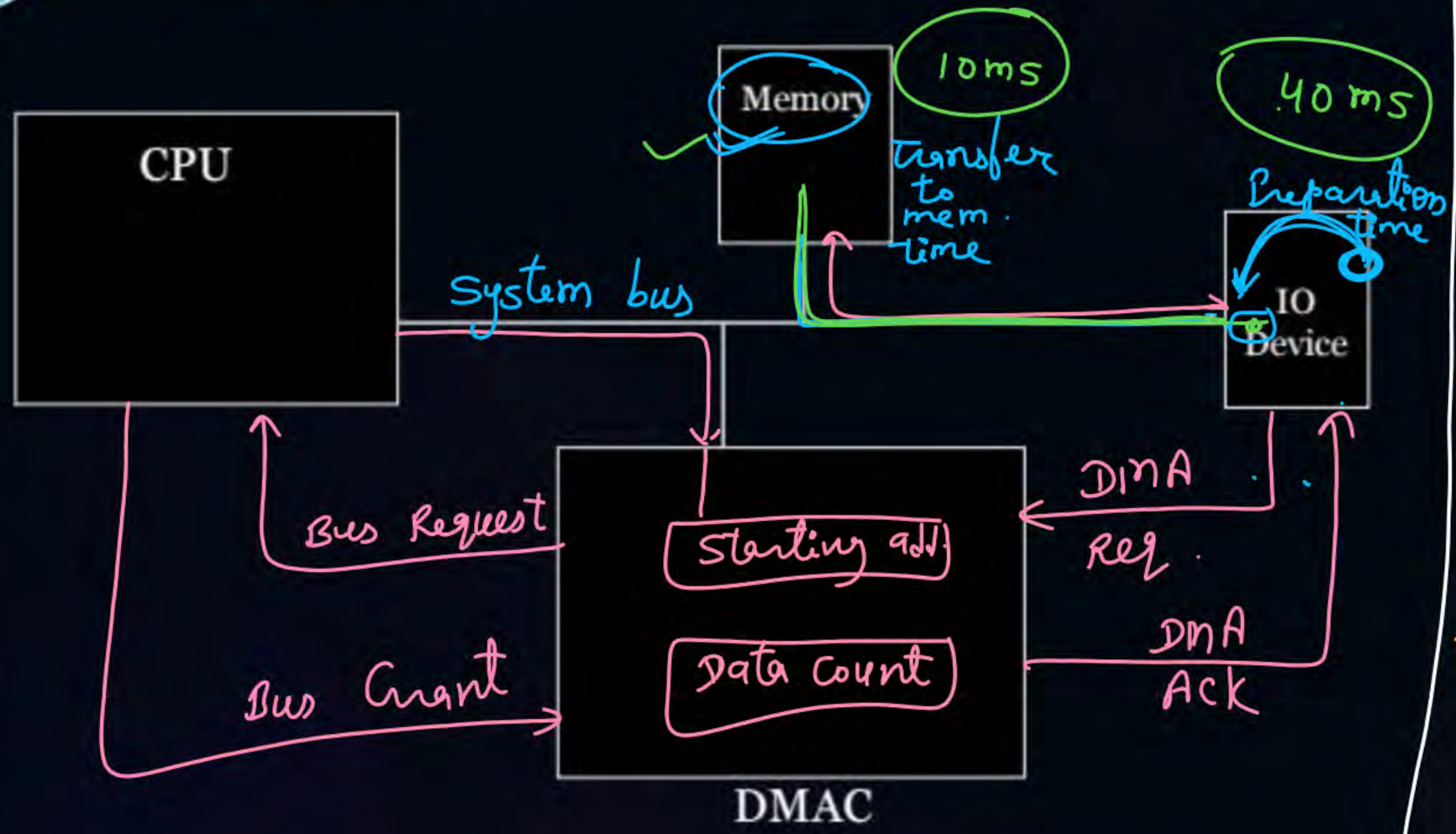


Topic DMA

Topic Modes of DMA



Topic : DMA

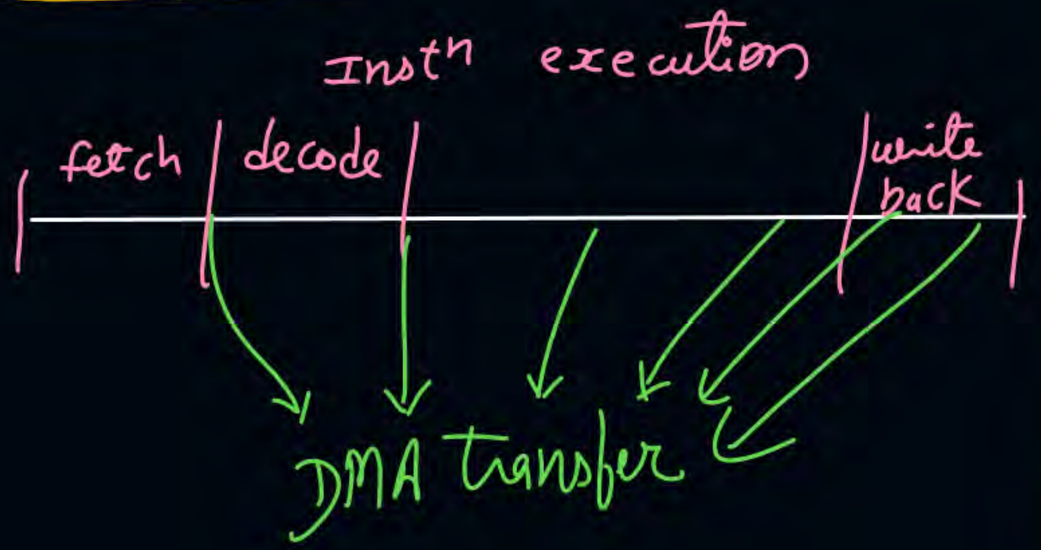


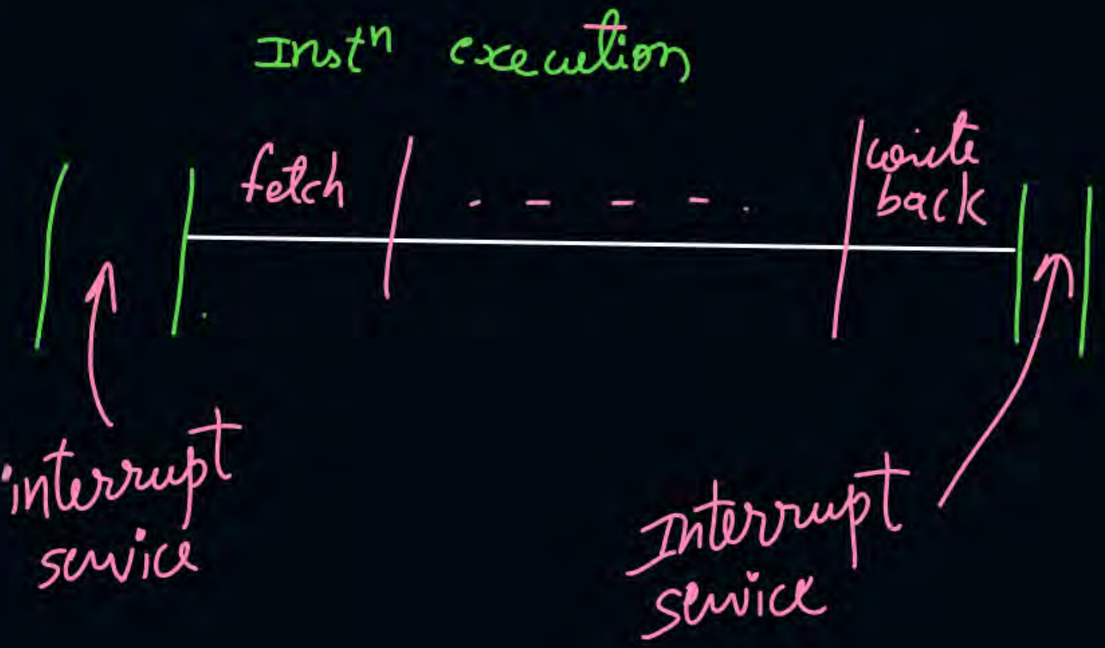
DMAC is a special purpose processor which is used for data transfer b/w mem. & I/O.

⇓

DMAC sends add. and control signals for this transfer.

→ when CPU gives control of the buses to DMAc then it's all register values (status of current program) will remain as it is inside CPU.







Topic : DMA

During DMA transfer, CPU can perform only those operations which do not require system buses.

which means CPU will be idle (blocked)

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for better user experience, control of the buses can not be given to DMAC for long time at a time.



Topic : Modes of DMA Transfer

when control of buses to be given to DMAC,
and if given then how much data to be transferred
by DMAC before CPU takes back the control of buses.



Topic : Modes of DMA Transfer

1. Burst Mode
2. Cycle Stealing
3. Interleaving DMA



Topic : Modes of DMA Transfer

Burst Mode :

DMAC transfers a block of data before CPU takes back control of the buses.

→ 512 B to 4KB



Topic : Modes of DMA Transfer

Cycle Stealing :

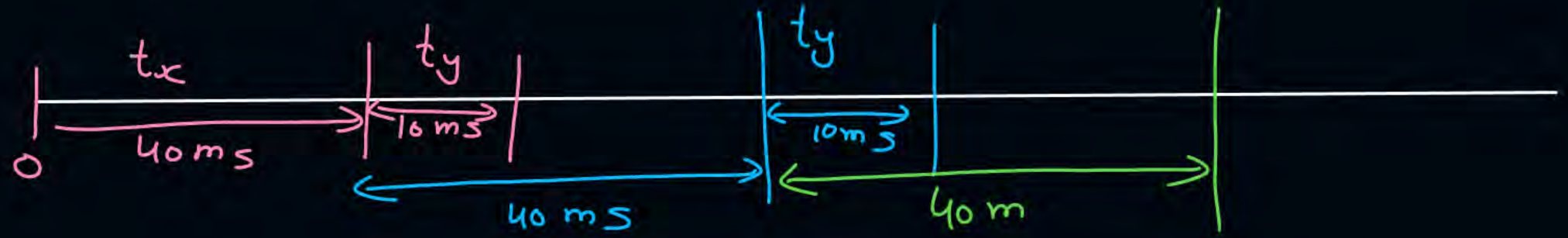
when slow I/O device prepares data then CPU keeps control of the buses. when data is ready then DMAC steals buses for sometime to transfer prepared data to memory.

Time required to prepare data in I/O = t_x

—11— transfer —11— to mem. = t_y

% of time CPU blocked due to DMA (burst mode) = $\frac{t_y}{t_x + t_y} * 100\%$

—11— (cycle stealing) = $\frac{t_y}{t_x} * 100\%$



$t_x = 40$ ms

$t_y = 10$ ms

t_x ⇒ based on speed of I/O device

t_y ⇒ —11— —11— memory.



Topic : Modes of DMA Transfer

Interleaving DMA:

CPU gives control of the buses to DMA, only when CPU does not need it.

⇒ Percentage of time CPU blocked due to DMA is almost Zero.

⇒ DMA transfer takes more time.

[MCQ]

Ans = 12.5

#Q. Consider a device operating on 1MBPS speed and transferring the data to memory using cycle stealing mode of DMA. If it takes 2 microseconds to transfer 16 bytes data to memory when it is ready/prepared. Then percentage of time CPU is blocked due to DMA is?

% of time CPU blocked due to DMA = $\frac{2\ \mu sec}{16\ \mu sec} * 100\%$

= 12.5 %

---

for 1MB, preparation time = 1sec

for 16B, —||— = $\frac{1sec}{1MB} * 16B = 16\ \mu sec$

[MCQ]

#Q. Consider a device operating on 5MBPS speed and transferring the data to memory using cycle stealing mode of DMA. If it takes x microseconds to transfer 8 bytes data to memory when it is ready/prepared. Then percentage of time CPU is blocked due to DMA is 25%. The value of x is 0.4 ?

$$25\% = \frac{x}{1.6\,\mu sec} * 100\%$$

$$x = \frac{1.6\,\mu sec}{4} = 0.4\,\mu sec$$

---

for 5 MB, time = 1 sec

for 8 B —11— = $\frac{1\,sec}{5\,MB} * 8B$

$= 1.6\,\mu sec$

[MCQ]

#Q. Consider a device transferring the data to memory using cycle stealing mode of DMA. If it takes 10 microseconds to transfer 8 bytes data to memory when it is ready/prepared and percentage of time CPU is blocked due to DMA is 10%. The data preparation time in the device is ___microseconds?

$$10\% = \frac{10\ \mu sec}{t_x} * 100\%$$

$$t_x = 100\ \mu sec$$

Ques) In prev question, speed of I/O is ?

in 100 μsec time, data prepared = 8 B

in 1 μsec —"— = $\frac{8\ B}{100\ \mu sec}$

in 1 sec —"— = $\frac{8\ B}{100 * 10^{-6} sec}$

= $\frac{8}{100}$ MB/sec

= 0.08 MB/sec

= 80 KB/sec

[NAT] GATE-2021

Ans = 80000

#Q. Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through cycle stealing at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is bits per second?

$t_y = 1 \text{ CPU cycle} = \frac{1}{2 MHz} = 0.5 \mu sec$

$0.5\% = \frac{0.5 \mu sec}{t_x} * 100\%$

$t_x = 100 \mu sec$

in 100 μsec, data prepared = 8 bits

in 1 sec —— 11 —— $= \frac{8 \text{ bits}}{100 * 10^{-6} \text{ sec}}$

$= 80000$ bits/sec

[NAT]

GATE H.W.

#Q. Consider a computer with a 4MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate (in bits per second) of the DMA controller if 1% of the processor cycles are used for DMA?

A 2,56,000

B 3,200

C 25,60,000

D 32,000

[MCQ]

#Q. A hard disk with a transfer rate of 10 Mbytes/second is constantly transferring data to memory using DMA. The processor runs at 600 MHz, and takes 300 and 900 clock cycles to initiate and complete DMA transfer respectively. If the size of the transfer is 20 Kbytes, what is the percentage of processor time consumed for the transfer operation ?

A 5.0%

B 1.0%

C 0.5%

D 0.1%

[NAT] P.Y.Q. H.W.

#Q. On a non-pipelined sequential processor, a program segment, which is the part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

```
        Initialize the address register
        Initialize the count to 500
LOOP:   Load a byte from device
        Store in memory at address given by address register
        Increment the address register
        Decrement the count
        If count !=0 go to LOOP
```

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.

The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.

What is the approximate speed up when the DMA controller based design is used in a place of the interrupt driven program based input-output?

[NAT]

#Q. The DMA controller has data count register of size 8-bits. The memory is byte addressable. The maximum number of bytes the DMA can transfer to memory at a time without giving the control of the buses back to CPU?

max value kept in data count = $(11111111)_2$

$= (255)_{10}$

Ans = 255 B

[NAT]

#Q. The DMA controller has data count register of size 8-bits. The memory is byte addressable.

1. Minimum how many times DMA needs to take control from CPU to transfer a file of 500 bytes? $\left\lceil \frac{500B}{255B} \right\rceil = 2$
2. Minimum how many times DMA needs to take control from CPU to transfer a file of 15K bytes?

$$\left\lceil \frac{15 * 1024\,B}{255\,B} \right\rceil = 61$$

[NAT]

GATE- 2016

#Q. The size of the data count register of a DMA controller is 16bits.

The processor needs to transfer a file of 29, 154 kilobytes from disk to main memory.

The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is 456 .

$$= \left\lceil \frac{29154 * 1024\,B}{(2^{16}-1)\,B} \right\rceil = 456$$



2 mins Summary

Topic DMA

Topic Modes of DMA



Happy Learning

THANK - YOU